Product Requirements Document: SankofaPOWER Platform

1. Executive Summary

Product Name: Sankofa
Type: Community-based resource sharing platform with geographic awareness
Core Value Proposition: A polylocal platform connecting individuals and institutions for resource exchange, knowledge sharing, and community building.

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2. User Types & Roles

2.1 Individual Users

Field	Type	Notes
Name	Text	Can be individual or institution name
Description	Text	"What do you do? What are you all about?"
Skills	Array	Including credentials
Attributes	Multi-select checkboxes	Growing, Culinary, Advocacy/Activist, Foraging, Educator/Knowledge Holder, Tools/Supplier, Food provider
Photo	Image	Profile picture
Cross Streets	Text	Internal use only - determines visibility radius
Census Tract	Auto-generated	Derived from cross streets
Contact Info	Text	Phone/Instagram/email (visible on account)
Notification Email/SMS	Text	For announcements and communications
Privacy Settings	Enum	Visibility based on physical distance to cross streets

2.2 Site Administrators

Individual users with elevated privileges:

- Delete users, sites, discussions, wikis (if inappropriate)
- Access all content regardless of user-set restrictions
- Contact all users, sites, discussions, wikis

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3. Modules

3.1 Community Map Module

3.1.1 Sites

Field	Type	Required	Notes
Name	Text	Yes	
Description	Text	Yes	Resources and offering available, e.g. controlled or uncontrolled food production (community garden, foraging site etc). Details of edible and medicinal plants if relevant. Soil health. Ongoing labor and material needs.
Creator	User ID	Yes	Link to user who created it
Additional Users	User IDs	No	Original user can add others
Open Hours	Time Range	No	Optional
Expiration Date	DateTime	No	For events/temporary sites
Accessibility	Text	No	Wheelchair access, gated, etc., membership process if relevant
Attributes	Multi-select	Yes	Same as user attributes
Contact Info	Text	Yes	
Reliability Score	Integer	System-generated	Up/down vote aggregate

3.1.2 Discussions

Field	Type	Required	Notes
Title	Text	Yes	
Question	Text	Yes	Open text field for discussion question/topic
Timestamp	DateTime	Yes	Auto-generated
Location	Geo-coordinates	Yes	
Expiration Date	DateTime	No	Optional
Reliability Score	Integer	System-generated	Up/down vote aggregate
Tag	Tag	No	Able to tag discussions to sites, users, wikis

3.1.3 Wikis (Place-Based Knowledge)

Field	Type	Required	Notes
Title	Text	Yes	
Content	Rich Text	Yes	
Location / Tag to site	Geo-coordinates	Yes	
Type	Enum	Yes	History/Story; Public Land Info; Knowledge-Sharing; Sacred/Ritual
Reliability Score	Integer	System-generated	Up/down vote aggregate

3.1.4 Ecological data

- Admin created only. Options including, for example (not exhaustive nor prescriptive):
    - Real time RSS feed of weather service data on that location
    - Soil health

    - Moon phases
    - Migratory patterns
    - Planting seasons
    - Heat maps: Real-time heat data, or historical averages
    - USDA hardiness zone and hyperlocal microclimate readings
    - Sourced from farmers almanac

3.1.5 Map View Specifications

- **Layers (toggleable):** Sites, Discussions, Wikis, Users, Ecological
    - **User Display:** Census tract outlines group users within the census tract
- **Filters:**
    - Site filters: Site designations/attributes
    - User filters: User designations/attributes
        - Same as site filters
    - Wiki filters: Wiki types
        - History/Story
        - Public Land Info
        - Knowledge-Sharing
        - Sacred/Ritual
    - Ecological: subtypes of ecological data
- **Colors:** Four distinct colors for each content type (Sites, Discussions, Wikis, Users)
    - Sites, Discussions, Wikis are points, Users are census tract outlines
    - **Overlapping Points:** Single point with color indicators + expandable list

3.1.6 List View Specifications

- Same filters as map view
- Sorting options:
    - Physical distance from selected point/address (default: user profile location)
    - Most recently added

3.1.7 Search Functionality

- Searches titles and descriptions across all content types

3.2 Tradepost Module

3.2.1 Postings Structure

Field	Type	Required	Notes
Type	Enum	Yes	Offer OR Ask

Creator	User ID	Yes	Expandable user details
Title	Text	Yes	
Description	Text	Yes	
Transaction Type	Enum	Yes	Free, Paid, Trade (multiple selectable except Free)
Quantity	Integer	Yes	With periodic reminders to update
Date Range	DateTime Pair	Yes	Start/End
Location	Geo-coordinates	Yes	Address or census tract display option
Tags	Array	Yes	User-generated OR pre-populated
Reliability Score	Integer	System-generated	Per-poster rating

3.2.2 Tag System

- User-generated option:
    - Users prompted with previously used tags
    - Option to create new tags
- Pre-populated option:
    - Only admins can create tags
- Tag subscriptions available
    - Notifications via contact info for subscribed tags

3.2.3 Tradepost Views

- **List View:** Offers/Asks/Sites with filters
- **Filters:** Minimum upvote %, Free/Paid/Trade, Tags
- **Sorting:** Recently added, Distance, Expiring soonest

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4. Technical Requirements

4.1 Geographic Visibility System

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None
Visibility Logic:
├── User Account Level
│   └── Privacy setting based on physical distance to cross streets
├── Content Level (Sites/Discussions/Wikis/Tradepost)
│   └── Independent visibility limits (may differ from user account
settings)
└── Implementation Note:
    └── Cross streets → Census tract auto-generation (internal
only)
```

4.2 Quality Control Mechanisms

- Up/down voting system for:
  - Site reliability (ex. does it do what it says?)
  - Wiki accuracy (ex. is information correct?)
  - Tradepost user reliability (ex. follow-through on offers/asks)
- Filter capability: Hide posters/users below rating threshold

4.3 Data Relationships

```
None
User ——creates——> Site
User ——creates——> Discussion
User ——creates——> Wiki
User ——creates——> Tradepost Posting
Tradepost Posting ——contains——> Tags
Tag ——subscribed_by——> User
Census Tract ——contains——> Users/Content
```

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5. Non-Functional Requirements

5.1 Performance

- Real-time location-based filtering
- Efficient geospatial queries for proximity sorting
- Map rendering optimization for overlapping points

5.2 Security & Privacy

- User-set visibility restrictions must be enforced server-side
- Admin override capabilities documented and logged
- Contact information visibility controlled by user settings

5.3 Scalability Considerations

- Census tract-based aggregation for user density visualization
- Tag subscription notification system at scale
- Rating system preventing manipulation

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6. Open Questions / Decisions Needed

#	Question	Impact
1	Security, Captcha, behavioral controls on accounts	Authentication, security
2	Are tags user-generated or pre-populated?	UI/UX, indexing
3	How many tags max per post? (1-3)	Validation rules
4	Real-time ecological data vs. pre-calculated averages?	API dependencies, infrastructure
5	Notification delivery method priority (email vs. SMS)?	Third-party service selection
6	Admin user creation process? (manual approval, invitation, automated?)	Authentication flow
7	Map icons: can/should different subtypes of sites and wikis have separate icon types?	Design
8	Is a map view necessary for the Tradepost?	Build

7. MVP Scope Recommendation

Phase 1 (Core):

- User registration with basic profile
- Site creation and map display
- Wikis
- Basic tradepost (offers/asks)
- Distance-based visibility
- Up/down voting system

Phase 2 (Enhanced):

- Discussions
- Tag subscription system
- Advanced filtering
- Admin dashboard
- Ecological data layer

Phase 3 (Advanced):

- Real-time environmental data integration
- SMS notifications
- Advanced analytics

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8. Next Steps for Sankofa Team

1. **Concept testing**: surveys and focus groups with potential users planned and scheduled for Spring 2026
2. **Wireframes:** Develop for map views, list views, profile pages

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9. User Journeys

9.1. The Local Business Owner

Goal: Transform a dirt patch next to the parking lot into a community garden. **Persona:** "GreenLeaf Market" owner.

- **Account set up**
  - GreenLeaf Market owner creates a **User account** with their personal name and home address/census tract.
  - The owner creates a **Site** for GreenLeaf Market, including relevant details.
- **Step 1: Discovery & Assessment**
  - The owner opens the **Map View** and toggles the **Ecological Layer** to check *(for example)* soil health data and sun exposure patterns for their specific block.
  - They search the map for existing "Growing" sites nearby to understand possible demand and resources.
- **Step 2: Offer Creation**
  - They create a new **Offer** titled "Small plot of land for community garden"
  - **Attributes:** Selects "Free," Quantity 1, no specific date range
  - **Details:** Adds a description, including dimensions, and their expectations for site maintenance. Includes phone number. Tags offer with `#Land #Garden`
  - **Visibility:** within a 3-mile radius of GreenLeaf Market.
  - A local gardener calls their phone number in the offer.
- **Step 3: Planning**
  - Neither the gardener nor the GreenLeaf Market owner know much about local zoning, so the owner starts a **Discussion** to ask the community what zoning laws might pertain to the particular garden space.
  - ALTERNATE: The owner posts an **Ask** in the tradepost, “Guidance on zoning navigation for a new community garden.” They select
  - A knowledgeable neighbor responds to the discussion/ask.
- **Step 3: Garden Creation & Engagement**
  - The gardener creates a new **Site** at the same location as GreenLeaf Market, called “GreenLeaf Community Garden.”
  - They decide to split the plot up and post an **Offer** in the **Tradepost**: “Plots in a new community garden,” for four gardeners to each cultivate a 10’x5’ area. Tags offer with `#GardenPlot`
  - They post an **Ask**: "Volunteers to help till a new community garden," selecting “Free” and setting a two-week date range in early May. Tags ask with `#GardenVolunteer`
  - They post another **Ask**: “Garden shed,” looking for materials or a pre-fab shed in exchange for a gift card to GreenLeaf Market. Tags ask with `#GardenSupplies`
- **Outcome:** The garden is built with volunteer labor, and the business gains more community visibility.

9.2. The New Gardener

Goal: Learn local growing tips and find a mentor. **Persona:** "Alex," a recent transplant to the neighborhood.

- **Step 1: Onboarding**
    - Alex creates an account, entering their cross streets (internal) and selecting attributes: "Growing"
    - They set their privacy to "Visible within 1 mile"
- **Step 2: Knowledge Retrieval**
    - Alex opens the **Map View**, toggles the **Wiki** layer, and filters by "Knowledge-Sharing."
    - They find a wiki entry: "Gardening Tips for Clay Soil in [Neighborhood Name]."
    - They read the content and upvote it for reliability.
- **Step 3: Mentorship Connection**
    - Alex switches to **List view,** toggles the **Users** layer, filters by "Educator/Knowledge Holder" attribute and sorts by "Distance."
    - The search yields several gardeners, and Alex navigates to one profile within their census tract and reaches out via provided contact information.
- **Step 4: Participation**
    - Alex navigates to the **Tradepost** and searches the word “garden.” They find an offer to join a local gardening club, and reach out via provided contact information.
- **Outcome:** Alex avoids common local soil mistakes and establishes mentorship relationships.

9.3. The Community Center

Goal: Promote a free weekend culinary education festival to the immediate neighborhood. **Persona:** "Riverside Community Center" coordinator.

- **Account set up**
    - Riverside Community Center coordinator creates a **User account** with their personal name and home address/census tract, and sets visibility to 0.5 miles.
    - The coordinator creates a **Site** for Riverside Community Center, including relevant details. Privacy is set to none.
        - Note: Anyone who lives outside of 0.5 miles around the coordinator’s personal residence will not see their account associated with the community center, however anyone can see the community center itself.
- **Step 1: Event Setup**
    - The coordinator creates a second **Site** at the same location as the Community Center called "Riverside Culinary Festival." They include relevant promotional details including a photo of the promotional poster and a link to register.

    - **Attributes:** Selects "Culinary" and "Educator."
    - **Expiration:** Sets an expiration date for the final day of the festival.
    - Since the event features direct benefits for participants (ex. learning opportunities), the coordinator also posts an **Offer** in the Tradepost: “Free cooking classes at Riverside Culinary Festival” selecting “Free,” and including the dates, a link to register, and # of available seats. They tag the post with `#CookingClasses`
- **Step 4: Feedback Loop**
    - After the class, participants upvote the **Site** and **Offer** for "reliability"
- **Outcome:** High attendance from the immediate neighborhood due to precise geo-targeting and trust signals.

9.4. The Mobile Food Vendor

Goal: Update customers on daily location changes. **Persona:** "Maria's Tacos."

- **Step 1: Dynamic Posting**
    - Maria logs in and creates a new **Offer** (or updates an existing recurring one).
    - **Title:** "Maria's Tacos - Today at [Park Name]." Selects “Paid” and “Trade.” Today, Maria decides she would also be willing to trade tacos for help cleaning her mobile food cart.
    - **Location:** Updates the geo-coordinates to the park's entrance.
    - **Date Range:** Sets start/end times for today only.
    - **Tags:** Uses `#MobileVendor`, `#Tacos`, and `#Lunch`.
    - Includes contact information
- **Step 2: Feedback Loop**
    - Users reach out to help with cleaning, and then upvote the **Site** for "reliability," increasing Maria’s reliability rating for future Offers.
- **Outcome:** Increased foot traffic at the new location, opportunity for bartering.

9.5. The Seasoned Gardener

Goal: Borrow a tiller and trade surplus tomatoes. **Persona:** "Sam," an experienced home gardener.

- **Step 1: Tradepost Search**
    - Sam filters the **Tradepost** by "Trade" and "Tools/Supplier."
    - He finds a neighbor with a "Power Tiller" listed for trade. He reaches out via the contact information listed and arranges a trade over the weekend for a flat of heirloom tomato seedlings.
- **Step 1 alt: Tradepost Posting**

    - Sam doesn’t find what he needs in **Offers**, so he creates an **Ask**: "Need Tiller for 2 days." He sets the two-week date range in which he is flexible to use it.
    - He selects "Trade” and indicates that he has a flat of heirloom tomato seedlings to offer in exchange.
    - A neighbor reaches out via Sam’s provided contact information.
- **Step 2: Trust Verification**
    - Before meeting up with the neighbor (either from the Offer or the Ask), Sam checks the neighbor's **Reliability Score** (up/down votes from previous trades). It is high.
- **Step 3: Completion & Rating**
    - After the trade, both parties upvote each other, reinforcing the community trust network.
- **Outcome:** Sam gets the heavy machinery he needs without buying it; the neighbor gets seedlings.

9.6. The Forager

Goal: Share walnut tree locations and discover new spots. **Persona:** "Elena," a local foraging enthusiast.

- **Step 1: Knowledge Contribution**
    - Elena creates a **Wiki** entry: "Abundant Black Walnuts on Oak Street."
    - **Type:** Selects "Public Land Info"
    - **Content:** Describes the tree's yield, best harvest time, and safety notes (allergies).
    - **Location:** Pins the exact GPS coordinates, sets visibility to 1 mile at first to prioritize letting neighbors know. She decides to change the visibility to 5 miles later on.
- **Step 2: Discovery**
    - She browses the **Map View**, toggling the **Wiki** layer and filtering for "Public Land Info."
    - She discovers a "Mushroom Spot" wiki created by another user, which she hadn't known about.
- **Step 3: Community Building**
    - She starts a **Discussion** titled "Walnut Harvest Party?" to organize a group gathering.
- **Outcome:** Local food resources are mapped and shared safely, preventing waste and building community connection.

9.7. The Policy Advocate

Goal: Organize neighbors to support a city council ordinance. **Persona:** "David," a local activist.

- **Step 1: Spreading the Word**
    - David creates an **Ask** called "Support Community Gardens Ordinance" to ask local gardeners to show up in support at the next city council meeting.
    - **Details**: Sets start and end date for a Tuesday in three weeks, when the city council will be meeting. Sets the location as City Hall. Number is unlimited. Time is listed in the description. Ask is set to “Free” with tags `#Policy #Garden`
    - **Visibility:** Sets to a 10-mile radius to make sure everyone in the city can see it.
    - He includes a link to a registration form so he can track who will be joining.
- **Step 2: Mobilization**
    - He creates a **Wiki** with the location of City Hall explaining the ordinance details and providing talking points.
- **Step 3: Engagement**
    - Neighbors start a **Discussion** also with the location City Hall (so that it shows up alongside the Wiki) to ask public questions about the ordinance.
    - David responds, as does the city council member who proposed the ordinance.
- **Outcome:** A coordinated group of neighbors attends the meeting, increasing political pressure.

9.8. The Neighbor In Need

Goal: Find reliable, free food sources without stigma. **Persona:** "Jordan," currently facing financial hardship.

- **Step 1: Tradepost Search**
    - Jordan searches the **Tradepost** for "Free" items and filters by "Food." Jordan verifies high reliability before reaching out to the contact info on a listed **Offer**.
- **Step 2: Map Search**
    - Jordan opens **List View**, toggles **Sites**, and filters by “Food provider” to find free or low-cost food resources. Jordan also searches "Food Shelves" and "Community Fridges" separately to find specific types of resources.
    - They check the **Reliability Score** of the site to ensure it is active and trustworthy.
- **Step 3: Feedback**
    - Jordan visits a food shelf during listed open hours, and meets a neighbor offering surplus garden tomatoes, returning to the app to upvote them both for reliability.
- **Step 4: Reciprocity (Optional)**
    - Later, Jordan sees the same gardener post an **Ask** for help with winterizing their garden. Jordan reaches back out via contact info and offers a hand.
- **Outcome:** Jordan accesses essential resources efficiently and privately, preserving their sense of agency.